Client Code –

```
#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/stat.h>
#include<unistd.h>
#include<fcntl.h>
#include<string.h>

int main()
{
   puts("\n\tClient - Listening\n");
   int code6 = mkfifo("fifo6.txt",0666);
   int code7 = mkfifo("fifo7.txt",0666);
   char strMessage[5000];
   if(code6 == -1)
      perror("\n\tmkfifo6 returned an error-file any already exist\n");
   if(code7 == -1)
      perror("\n\tmkfifo7 returned an error-file any already exist\n");

   int fd = open("fifo6.txt", O_RDONLY);
   int fd2 = open("fifo7.txt", O_WRONLY);
   if(fd == -1)
   {
      perror("Cannot open FIFO6 for read");
      return EXIT_FAILURE;
   }
   if(fd2 == -1)
   {
      perror("Cannot open FIFO7 for write");
```

```
    return EXIT_FAILURE;
}
puts("FIFO OPEN");
//read string up to(5000 characters)
char stringBuffer[5000];
memset(stringBuffer, 0, 5000);

int res;
char Len;
//while(1)
{
    res = read(fd, &Len, 1);
    //if(Len == 1)//since null counts 1
        //break;

    read(fd, stringBuffer, Len); //Read String Characters
    stringBuffer[(int)Len] = 0;
    printf("\nClient Received: %s\n", stringBuffer);
    int j = 0,w=0, line = 0;
    while(stringBuffer[j]!='\0'){
        char ch = stringBuffer[j];
        if((ch==' ')||(ch=='\n')){
            w++;
            if(ch=='\n')
                line++;
        }
        j++;
    }
    char LC = (char) strlen(strMessage);
    char str1[256];
    char str2[256];
```

```
    char str3[256];
    sprintf(str1," No.of Words : %d:::", w); strcat(strMessage,str1);
    sprintf(str2," No.of Charecters: %d:::",(j-1)); strcat(strMessage,str2);
    sprintf(str3," No.of Lines: %d",line); strcat(strMessage,str3);
    strcat(strMessage,"\0");
       printf("\n\tString: %s",strMessage);
    write(fd2, &LC, 1);
    write(fd2, strMessage, strlen(strMessage));
    fflush(stdin);
    strMessage[0] = 0;//reseting the character array
    //if(LC==1)
       //break;

  }
  printf("\n");
  puts("CLIENT CLOSED");
  puts("SERVER CLOSED");
  close(fd);
  close(fd2);
  return 0;
}
```

Server Code –

```
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<sys/types.h>
#include<sys/types.h>
#include<fcntl.h>
#include<string.h>
```

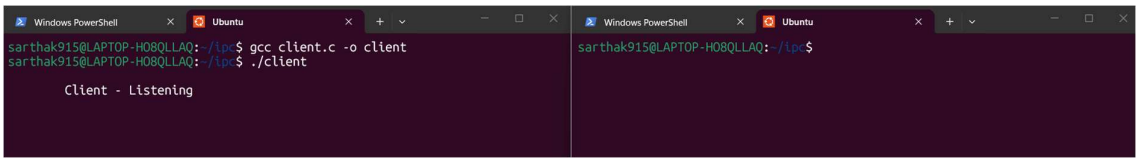
```
int main()
{
   int n;
   puts("Server");
   char strMessage[5000];//[] = {"welcome", "to", "the", "module.", "This", "will", "now", "stop"};
   int fd = open("fifo6.txt", O_WRONLY);
   int fd2 = open ("fifo7.txt", O_RDONLY);
   if(fd == -1)
   {
      perror("cannot open fifo6");
      return EXIT_FAILURE;
   }
   if(fd2 == -1)
   {
      perror("cannot open fifo7");
      return EXIT_FAILURE;
   }
   puts("FIFO OPEN");
   //read string up to(5000 characters)
   char stringBuffer[5000];
   memset(stringBuffer, 0, 5000);
   int res;
   char Len;
   //while(1)
   {
      printf("\n\n\t\tEnter the Message to be passed (hitting ENTER without any string will
terminate program): ");
      fgets(strMessage, 100, stdin);
      char L = (char) strlen(strMessage);
      //printf("\n\tLength of the given string: %d\n", (L-1));
```

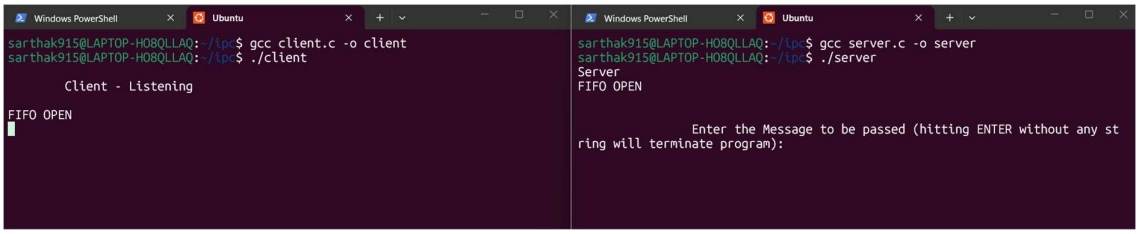
```
        write(fd, &L, 1);
        write(fd, strMessage, strlen(strMessage));
        fflush(stdin);
        strMessage[0] = 0;//reseting the character array
        //if(L==1)//since null counts 1
           //break;
        int len2;
        res = read(fd2, &len2, 1);
        //if(len2 == 1)//since null counts 1
           //break;

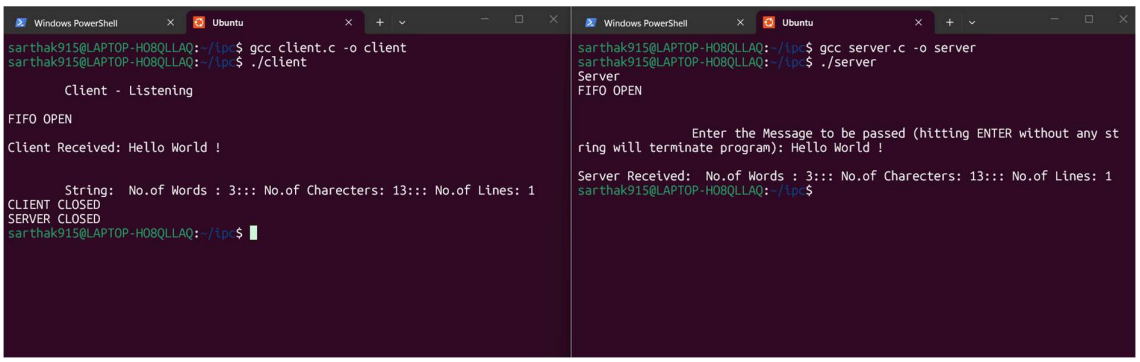
        read(fd2, stringBuffer, 5000); //Read String Characters

        printf("\nServer Received: %s\n", stringBuffer);
        stringBuffer[(int)len2] = 0;
    };
    //printf("\n\nCLIENT CLOSED\n")
    //return 0;
}
```

OUTPUT -


Windows PowerShell
Ubuntu
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ gcc client.c -o client
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ ./client
Client - Listening
Windows PowerShell
Ubuntu
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$


Windows PowerShell
Ubuntu
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ gcc client.c -o client
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ ./client
Client - Listening
FIFO OPEN
Windows PowerShell
Ubuntu
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ gcc server.c -o server
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ ./server
Server
FIFO OPEN
Enter the Message to be passed (hitting ENTER without any st
ring will terminate program):


Windows PowerShell
Ubuntu
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ gcc client.c -o client
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ ./client
Client - Listening
FIFO OPEN
Client Received: Hello World !
String: No.of Words : 3::: No.of Charecters: 13::: No.of Lines: 1
CLIENT CLOSED
SERVER CLOSED
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$
Windows PowerShell
Ubuntu
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ gcc server.c -o server
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$ ./server
Server
FIFO OPEN
Enter the Message to be passed (hitting ENTER without any st
ring will terminate program): Hello World !
Server Received: No.of Words : 3::: No.of Charecters: 13::: No.of Lines: 1
sarthak915@LAPTOP-HO8QLLAQ:~/ipc$